PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES (W2-PM-FINAL)

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Program/Batch: B084-Networkwalks

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Modules Completed: W2-PM1 & W2-PM5

Permission Secured: Yes (Written Consent)

Client / Target: 1. Networkwalks (Authorized)
2. Local LAN Network

Domain: Cybersecurity & Ethical Hacking

Phases Covered: Phase 1: Reconnaissance & Footprinting | Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

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2. Introduction

This report covers footprinting the `networkwalks.com` domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). Module W2-PM1 focuses on the reconnaissance and footprinting phase, while W2-PM5 covers network discovery and mapping live hosts. Together, they demonstrate how an attacker transitions from passive public information gathering to active network enumeration.

All terminal commands were executed inside a Kali Linux virtual environment, and network discovery scans were conducted on a local LAN subnet. Each section documents the exact command utilized, observed outputs, evidence screenshots, and an analysis of the findings from an offensive security perspective.

3. Tools Used

Tool	Purpose
Kali Linux & Windows	Operating environments utilized for reconnaissance, scanning, and analysis.

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WHOIS	Retrieves domain registration records, creation dates, registrar details, and name servers.
WhatWeb	Identifies web server software, Content Management Systems (CMS), plugins, and frameworks.
nslookup	Resolves domain names to IP addresses via designated DNS servers.
curl -I	Inspects HTTP response headers, server banners, cookies, and security policies.
wafw00f	Detects and fingerprints Web Application Firewalls (WAF) protecting the target web app.
dnsrecon	Performs comprehensive DNS enumeration (NS, MX, SPF, TXT, SRV records).
Zenmap (Nmap GUI)	Scans local subnets to discover live hosts, IP addresses, MAC addresses, and topology.

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

I performed comprehensive reconnaissance against the target domain `networkwalks.com` using six specialized Kali Linux utilities:

- **WHOIS:** Queried domain registration records. The output confirmed registrar details (GoDaddy), creation date (2019-11-06), expiry date (2027-11-06), and HostGator name servers (`NS6135.HOSTGATOR.COM` and `NS6136.HOSTGATOR.COM`).
- **WhatWeb:** Fingerprinted web application technologies. The tool identified Apache HTTP Server, WordPress (v7.1.4), WordPress Download Manager (v3.3.76), jQuery (v3.7.1), and Bootstrap (v7.1.4) running on IP `192.232.216.135` in the United States.
- **nslookup:** Resolved `networkwalks.com` using Google's public DNS (`8.8.8.8`), successfully returning the target server IP address: `192.232.216.135`.
- **curl -I:** Inspected HTTP response headers. The request exposed critical backend indicators including Apache server headers, WordPress cookies (`__wpdm_client`), and REST API endpoints (`/wp-json/`).
- **wafw00f:** Probed for Web Application Firewalls. The scan successfully detected that `networkwalks.com` is protected behind a **ModSecurity (SpiderLabs) WAF**.
- **dnsrecon:** Enumerated complete DNS records. The utility identified SOA, NS, A, MX, TXT (including Google site verification and SPF records), and multiple SRV autodiscover records.

4.2 Network Scanning with Zenmap (W2-PM5)

For the second phase, I performed internal network discovery on my local subnet using Zenmap. After identifying the local interface configuration, I targeted the subnet with a Ping Scan (`nmap -sn`) to map live devices, extract IP and MAC addresses, and generate visual network topology diagrams.

5. Risk Analysis / Impact

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web Technology Information Exposed	WhatWeb identified WordPress v7.1.4 and WP Download Manager v3.3.76	Attackers can cross-reference exact software versions with known vulnerabilities (CVEs) for targeted exploitation.	Medium
2	Server IP Address Identifiable	Nslookup resolved domain to 192.232.216.135	Reveals the direct server hosting location, enabling potential direct-to-IP attacks or perimeter probing.	Low
3				Low

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
	HTTP Header Technical Disclosure	Curl returned headers exposing REST API endpoints (/wp-json/)	Aids in API enumeration and discovering administrative user routes or content endpoints.	
4	WAF Fingerprinting Visible	Wafw00f identified ModSecurity (SpiderLabs)	Informs attackers of defensive measures in place, allowing them to tailor evasion and bypass techniques.	**Low**
5	DNS Infrastructure Profiling	DNSRecon extracted NS, MX, TXT, and SRV records	Exposes mail server infrastructure and domain configuration details for social engineering or mail spoofing.	**Medium**
6	Active Devices Visible on LAN	Zenmap ping scan identified live hosts and MAC addresses	Reveals internal network layout and potential unauthorized or rogue devices connected to the local network.	**Medium**

Note: These observations represent information-gathering results and are not confirmed vulnerabilities. No exploitation or intrusive vulnerability testing was performed.

6. Recommendations

1. **Review Publicly Exposed Technology Information:** Audit what CMS banners, version strings, and plugin signatures are publicly broadcasted.
2. **Maintain Strict Software Patch Management:** Keep core CMS platforms, themes, and plugins updated against active security advisories.
3. **Harden HTTP Response Headers:** Remove unnecessary server banners, technology signatures, and debug endpoints from HTTP headers.
4. **Minimize DNS Record Exposure:** Restrict DNS zone transfers and ensure only necessary service records are publicly accessible.
5. **Tune and Monitor WAF Defenses:** Ensure ModSecurity rulesets are actively updated and configured to log and block malicious request anomalies.
6. **Conduct Regular Internal Network Audits:** Periodically perform internal host discovery scans to maintain asset inventory and visibility.
7. **Investigate Rogue Devices:** Promptly inspect and verify any unrecognized hardware or IP addresses discovered on the local network.
8. **Maintain Accurate Network Documentation:** Keep architecture diagrams, subnet maps, and device inventories updated.
9. **Enforce Strict Authorization Policies:** Ensure all reconnaissance and scanning activities are strictly governed by prior written consent and scope agreements.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks, I successfully executed practical exercises covering domain footprinting and network discovery. Using specialized Kali Linux reconnaissance tools, I gathered actionable intelligence regarding target infrastructure, DNS records, web technologies, and firewall protections. Furthermore, I utilized Zenmap to map active local network hosts and understand internal host discovery mechanics.

This engagement reinforced that comprehensive reconnaissance is a foundational pillar of ethical hacking. Security professionals must understand how external footprints can be leveraged by malicious actors to construct attack surfaces. All activities were successfully completed in strict accordance with educational authorization guidelines.

8. Evidences Collected

Below are the captured terminal outputs and tool interfaces serving as verifiable evidence for each completed lab module:

WHOIS Evidence

Figure 8.1: WHOIS command output showing domain registration details and name servers.

WhatWeb & Nslookup Evidence

Figure 8.2: WhatWeb fingerprinting and Nslookup IP resolution for networkwalks.com.

Curl Header Inspection

Figure 8.3: Curl -I inspecting HTTP response headers and REST API endpoints.

Wafw00f WAF Detection

Figure 8.4: Wafw00f identifying ModSecurity (SpiderLabs) WAF protection.

DNSRecon Enumeration

Figure 8.5: DNSRecon enumerating comprehensive DNS, MX, TXT, and SRV records.

Zenmap Network Discovery

Figure 8.6: Zenmap ping scan and topology mapping of local subnet active hosts.

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📌 **Project Information:** Program Name: Cybersecurity & Ethical Hacking Internship at Networkwalks | Batch: B084 | Week: 02 | Repository: GitHub